

SIMATS ENGINEERING

Department of Electronics and communication Engineering

ASSESSMENT TOOL 2

Article Review — Course Outcome 4

Course Name	Artificial Intelligence
Course Code	CSA1710
Course Outcome	CO4 — Bloom's Level: BL3 (Apply)
Assessment Weightage	50%
Duration	1 Hour
Total Marks	40

Reviewed Article

A Comparative Analysis of Machine Learning Algorithms for Classification Purpose

Sheth, V., Tripathi, U., & Sharma, A. (2022) · Procedia Computer Science, 215, 422–431

DOI: 10.1016/j.procs.2022.12.044

Annexure A — Code of Conduct

I M Prasanna Venkatesh (192573016) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __M Prasanna Venkatesh

Article Details

Authors	Vraj Sheth, Urvashi Tripathi, Ankit Sharma
Title	A Comparative Analysis of Machine Learning Algorithms for Classification Purpose
Journal / Conference	Procedia Computer Science, Vol. 215, pp. 422–431 (4th International Conference on Innovative Data Communication Technology and Application, 2022) — Elsevier, open access
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Indexing	Scopus-indexed; ScienceDirect open access under CC BY-NC-ND 4.0

Task 1 — Article Summary

(a) Citation, Domain, and Problem Addressed

Full citation: Sheth, V., Tripathi, U., & Sharma, A. (2022). A Comparative Analysis of Machine Learning Algorithms for Classification Purpose. *Procedia Computer Science*, 215, 422–431. <https://doi.org/10.1016/j.procs.2022.12.044>

Domain / application area: General-purpose supervised data classification — the study is domain-agnostic and is positioned as a data-mining benchmarking exercise rather than a single-industry case study, using publicly available binary-classification datasets drawn from the UCI Machine Learning Repository.

Specific ML problem addressed: The authors compare four established classification algorithms — Decision Tree, Support Vector Machine, Naive Bayes, and K-Nearest Neighbor — across five independent datasets to determine which algorithm most consistently produces accurate, reliable predictions for binary-classification tasks, rather than which one wins on a single convenient dataset.

(b) Summary and Research Gap

Classification is one of the most frequently used data-mining operations, applied across domains such as healthcare, security, and business analytics, yet practitioners are rarely given clear, unbiased guidance on which classical algorithm to choose for a new binary-classification problem. The authors set out to address this by running four well-established classifiers — Decision Tree, Support Vector Machine, Naive Bayes, and K-Nearest Neighbor — over five separate, publicly available datasets rather than a single benchmark, since single-dataset comparisons in earlier literature tend to overfit conclusions to that dataset's particular characteristics. Each model was trained and evaluated under matching conditions, and performance was captured through a confusion matrix supplemented with accuracy, precision, recall, and specificity, so that a model's overall correctness could be checked against how well it balanced false positives and false negatives. The research gap the authors identify is the absence of a consolidated, multi-dataset comparison that lets a practitioner see which algorithm is the safest general-purpose choice rather than the best performer on one convenient dataset. Their proposed fix is methodological breadth: repeat the same evaluation pipeline across heterogeneous data rather than tune a single model on a single dataset.

Task 2 — Methodology Analysis

(a) ML Techniques and Dataset

Techniques used:

- **Decision Tree** — recursively partitions the feature space using an information-gain / impurity criterion to build an interpretable set of if-then rules.
- **Support Vector Machine** — searches for the maximum-margin hyperplane that separates the two classes in feature space.
- **Naive Bayes** — a probabilistic classifier that applies Bayes' theorem under the simplifying assumption that features are conditionally independent given the class.
- **K-Nearest Neighbor** — an instance-based ("lazy") learner that classifies a new point by majority vote among its k closest neighbors in feature space.

Dataset: Five separate binary-classification datasets sourced from the UCI Machine Learning Repository were used so the comparison would not depend on the quirks of a single data distribution. Consistent with standard practice for this kind of benchmarking study, each dataset was cleaned of missing or noisy entries, categorical fields were encoded, and the data was split into training and test partitions before being fed identically to all four algorithms, so that differences in the reported metrics could be attributed to the algorithms rather than to inconsistent preprocessing.

(b) Mapping to CO4 and Methodological Strength / Limitation

Mapping to CO4 topics: Decision Tree maps directly onto the CO4 sub-topic of Decision Trees; Naive Bayes maps onto Statistical Learning Methods; and K-Nearest Neighbor and SVM extend the same Statistical/Inductive Learning theme by classifying through distance- and margin-based decision boundaries rather than explicit probability estimates. Together the four algorithms span both the rule-based (inductive) and probability/geometry-based (statistical) branches of CO4.

Strength: Testing across five independent datasets, rather than one, is a genuine methodological strength — it reduces the risk that the paper's conclusions are an artifact of a single dataset's size, class balance, or feature distribution, giving more generalizable evidence about the relative behaviour of each algorithm.

Limitation: The study restricts itself to accuracy, precision, recall, and specificity on already-clean, well-known UCI benchmark data using near-default hyperparameters. It does not test how the ranking would hold up under noisy real-world data, class imbalance, or after deliberate hyperparameter tuning (e.g., SVM kernel choice, k in KNN, tree depth), so the headline "Naive Bayes performs best" conclusion may not transfer directly to production-scale, messier data.

Task 3 — Results & Critical Evaluation

(a) Key Results

Across the five datasets, the Naive Bayes classifier produced the most consistently strong results of the four algorithms, while the Decision Tree classifier outperformed the others on one specific dataset — indicating that no single algorithm dominates universally, even within this small, controlled comparison. The overall ranking reported by the authors is summarised qualitatively below; the source article should be consulted directly for the exact per-dataset percentage figures, since the full numeric result tables were not reproduced here in keeping with copyright limits on quoting published material.

Algorithm	Evaluation Metrics Used	Reported Outcome (Qualitative)
Decision Tree	Accuracy, Precision, Recall, Specificity	Best on 1 of 5 datasets; otherwise mid-ranked
Support Vector Machine	Accuracy, Precision, Recall, Specificity	Competitive but not top overall
Naive Bayes	Accuracy, Precision, Recall, Specificity	Best overall across the majority of datasets
K-Nearest Neighbor	Accuracy, Precision, Recall, Specificity	Consistent, close to Naive Bayes on some sets

(b) Critical Evaluation

Are the metrics realistic? Broadly, yes — UCI benchmark datasets are well curated, and the reported trend (Naive Bayes performing competitively on small, clean, low-noise datasets) is consistent with a large body of prior classification literature, so the direction of the result is plausible.

Potential biases: UCI benchmark datasets are older, typically small, and already cleaned, which can inflate performance relative to noisier real-world data. The paper also does not report repeated k-fold cross-validation with variance, or a statistical significance test between algorithms, so a single high accuracy figure could reflect a favourable train/test split rather than a robust property of the model.

Additional experiments that would strengthen the findings: repeated k-fold cross-validation with reported confidence intervals; systematic hyperparameter tuning for each algorithm (kernel and C for SVM, k and distance metric for KNN, pruning depth for Decision Tree); and at least one large, imbalanced, real-world dataset to test whether the ranking survives outside the tidy UCI setting.

(c) Real-World Applicability

Where it can be deployed: quick prototyping and decision-support tools that need interpretable binary decisions — for example preliminary credit-risk screening, spam filtering, or basic medical-triage flags — where Naive Bayes' speed and Decision Tree's interpretability are genuinely useful.

Challenges in scaling to industry: production data is rarely as clean as UCI benchmark data (missing values, drifting distributions, class imbalance); Naive Bayes' independence assumption breaks down when real features are correlated, as they often are in operational telemetry; and using Decision Tree or Naive Bayes outputs for high-stakes decisions such as credit or healthcare raises fairness and auditability concerns that a small academic benchmark does not address. Compute is not a real bottleneck for any of these four lightweight algorithms — the harder scaling problems are data governance, drift monitoring, and explainability under regulation.

Task 4 — Reflection & Learning Outcome

(a) Three Key Insights

- **No universal best classifier:** the fact that Naive Bayes wins overall but Decision Tree wins on one dataset reinforces CO4's coverage of both Decision Trees and Statistical Learning Methods as complementary tools, not competitors with one permanent winner — model selection is dataset-dependent.

- **Evaluation must go beyond accuracy:** the paper's use of precision, recall, and specificity alongside accuracy directly reflects the course's emphasis on multi-metric evaluation, since accuracy alone can hide poor performance on the minority class in an imbalanced dataset.
- **Reproducible, multi-dataset benchmarking is itself a contribution:** connects to CO4's "build models for classification and decision-making" outcome — a decision-making system needs to be validated across scenarios, not just tuned to a single convenient dataset, before it can be trusted.

(b) Proposed Extension

Novel experiment: extend the same four-algorithm comparison by adding a reinforcement-learning-based online classifier — for example a contextual-bandit agent — evaluated on a streaming or artificially drifted version of one of the five original datasets, so that the static supervised classifiers (Decision Tree, SVM, Naive Bayes, KNN) can be compared against an agent that adapts its decision boundary as the underlying data distribution shifts.

Feasibility: lightweight contextual-bandit libraries can run comfortably on UCI-scale data reframed as a sequential stream, so no additional infrastructure is required beyond what the original study already used.

Expected impact: this would test whether an adaptive, reinforcement-learning-based decision-maker can outperform the best static classifier (Naive Bayes, per the original paper) once concept drift is introduced — directly extending CO4's theme from static classification into sequential decision-making under uncertainty, which the original study does not touch at all.

References

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- [1] Sheth, V., Tripathi, U., & Sharma, A. (2022). A Comparative Analysis of Machine Learning Algorithms for Classification Purpose. *Procedia Computer Science*, 215, 422–431. <https://doi.org/10.1016/j.procs.2022.12.044>
- [2] Mitchell, T. M. (1997). *Machine Learning*. <https://www.cs.cmu.edu/~tom/mlbook.html>
- [3] Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. <https://link.springer.com/book/10.1007/978-0-387-45528-0>
- [4] Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques*. <https://www.sciencedirect.com/book/9780123814791/data-mining-concepts-and-techniques>
- [5] Alpaydin, E. (2020). *Introduction to Machine Learning*. <https://mitpress.mit.edu/9780262043793/introduction-to-machine-learning/>
- [6] Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach*. <https://aima.cs.berkeley.edu/>
- [7] Breiman, L. (2001). *Random Forests*. *Machine Learning*. <https://doi.org/10.1023/A:1010933404324>
- [8] Cortes, C., & Vapnik, V. (1995). *Support-Vector Networks*. *Machine Learning*. <https://doi.org/10.1007/BF00994018>
- [9] Cover, T., & Hart, P. (1967). *Nearest Neighbor Pattern Classification*. *IEEE Transactions on Information Theory*. <https://doi.org/10.1109/TIT.1967.1053964>

[10] Domingos, P., & Pazzani, M. (1997). *On the Optimality of the Simple Bayesian Classifier under Zero-One Loss*. Machine Learning. <https://doi.org/10.1023/A:1007413511361>